



QUIZ 1

Course : Computational Statistics
Lecturer : Muhammad Afif Hendrawan, S.Kom., M.T.
Duration : 150 Minutes
Type : Open Book

In this QUIZ 1, we will analyze the WORLD HAPPINESS REPORT data. Please download the data from the page <https://p-3m.com/statkomkuis1>.

Quiz Rules:

- **ATTENTION!** The data is in comma-separated value (CSV) format. You may convert it to Microsoft Excel format if needed.
- You are only allowed to use **Microsoft Excel**. You may use either the desktop version or the cloud version (Office 365).
- **USING ANY COMMUNICATION TOOLS DURING THE QUIZ IS STRICTLY PROHIBITED.**

Case Study: Analyzing Global Happiness Trends (2019)

The World Happiness Report ranks countries based on their happiness scores, incorporating various factors such as GDP per capita, social support, and life expectancy. As a data analyst, your task is to explore this dataset to understand global happiness trends using statistical measures and visualizations.

Understanding Variables

1. Identify which variables in the dataset are categorical (ordinal/nominal) and which are numerical (discrete/continue). Also, identify the function of variables (independent/dependent). **(10 point)**

Central Tendency

1. Compute mean, median, and mode for all numerical variables. **(5 point)**
2. Compare the mean and median values. What does this indicate about the distribution of happiness scores and the distribution of generosity? Please explain it! **(5 point)**

Variability of Data

1. Compute range, variance, and standard deviation of score and generosity. **(5 point)**
2. Based on the standard deviation, describe whether happiness scores and generosity are **closely clustered** or **widely spread**. **(5 point)**
3. Compute Q1, Q2, Q3, and IQR from score and generosity. **(5 point)**



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4. Are there any outliers in the score" and generosity variables? If yes, state the total number of outliers, as well as the number of outliers exceeding the maximum and below the minimum. **(10 point)**

Data Visualization

1. Create a **bar chart** showing the top 10 happiest countries based on their happiness scores. **(10 point)**
2. Create a **pie chart** representing the proportion of countries in different happiness score using following ranges,
 - a. Low (score ≤ 4)
 - b. Medium ($4 < \text{score} \leq 6$)
 - c. High ($6 < \text{score} \leq 8$)
 - d. Very High ($8 < \text{score} \leq 10$)**(10 point)**
3. Create a **histogram** to visualize the distribution of happiness scores across all countries. **(15 point)**
4. Create a **boxplot** for score and generosity. Does the result of the box plot align with the findings in question no. 4 on variability of data section? Explain! **(20 point)**

BONUS: Correlation Between Variables

1. Create a scatter plot between score and generosity as well as between score and social Support. **(10 point)**
2. Explain the correlation between variables based on previous scatter plot. **(10 point)**